MILITARY SPECIFICATION

PLASTIC MOLDING AND EXTRUSION MATERIALS, POLYETHERSULFONE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers polyethersulfone (PES) thermoplastic materials suitable for injection molding or extrusion.

1.2 Classification. The PES thermoplastic materials shall be of the following types and classes (see 6.2):

Type I - Nonreinforced
- Class 1 - Injection molding resin
  - Grade A Melt flow = 43 ± 4 grams/10 min
  - Grade B Melt flow = 14.5 ± 1.5 grams/10 min
- Class 2 - Extrusion resin
  - Grade A Melt flow = 3.4 ± 0.4 grams/10 min

Type II - Glass fiber reinforced (GFR)
- Class 1 - 15% GFR
- Class 2 - 20% GFR
- Class 3 - 30% GFR
- Class 4 - 40% GFR
- Class 5 - 15% PTFE/30% GFR

Type III - Carbon fiber reinforced (CFR)
- Class 1 - 30% CFR
- Class 2 - 15% PTFE/30% CFR

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Director, US Army Materials and Mechanics Research Center, ATTN: DRXMR-SMS, Watertown, MA 02172 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

PPP-D-729 Drums: Metal, 55-gallon (For Shipment of Noncorrosive Material)

STANDARDS

MILITARY

MIL-STD-129 Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 149	Dielectric Breakdown Voltage and Dielectric Strength of Electrical Insulating Materials at Commercial Power Frequencies
ASTM D 150	A-C Loss Characteristics and Permittivity (Dielectric Constant) of Solid Electrical Insulating Materials
ASTM D 256	Impact Resistance of Plastics and Electrical Insulating Materials
ASTM D 257	D-C Resistance or Conductance of Insulating Materials
ASTM D 570	Water Absorption of Plastics
ASTM D 638	Tensile Properties of Plastics
ASTM D 648	Deflection Temperature of Plastics Under Flexural Load
ASTM D 696	Coefficient of Linear Thermal Expansion of Plastics
ASTM D 785	Rockwell Hardness of Plastics and Electrical Insulating Materials
ASTM D 790	Flexural Properties of Plastics and Electrical Insulating Materials
ASTM D 792	Specific Gravity and Density of Plastics by Displacement
ASTM D 955	Measuring Shrinkage From Mold Dimensions of Molded Plastics
ASTM D 1238	Flow Rates of Thermoplastics by Extrusion Plastometer
ASTM D 2863	Measuring the Minimum Oxygen Concentration to Support Candle-Like Combustion of Plastics (Oxygen Index)
ASTM D 3029	Impact Resistance of Rigid Plastic Sheeting or Parts by Means of a Tup (Falling Weight)
ASTM D 3951	Commercial Packaging

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ASTM D 3171 Fiber Content of Resin-Matrix Composites by Matrix Digestion
ASTM D 3418 Transition Temperatures of Polymers by Thermal Analysis

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

UNDERWRITERS LABORATORIES

UL 94 Tests for Flammability of Plastic Materials for Parts in Devices and Appliances
UL 746 Polymeric Materials - Long Term Property Evaluations

(Applications for copies of UL publications should be addressed to Underwriters Laboratories, Inc., Publication Stock, 333 Pfingsten Road, Northbrook, IL 60062.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Material. The material shall consist of polyethersulfone thermoplastic molding or extrusion material, with or without fiber reinforcement or PTFE lubricant (see 6.3).

3.2 Properties of PES thermoplastic. Unless otherwise specified (see 6.2), the properties of the PES thermoplastic materials shall be as specified herein.

3.2.1 Nonreinforced PES. The properties of the nonreinforced PES thermoplastic molding and extrusion materials shall be as specified in table I.

3.2.2 Glass fiber reinforced PES. The properties of the glass fiber reinforced PES thermoplastic materials shall be as specified in table II.

3.2.3 Carbon fiber reinforced PES. The properties of the carbon fiber reinforced PES thermoplastic materials shall be as specified in table III.

3.3 Form. The form of the PES thermoplastic resins shall be granules, pellets, or powder. When specified by the procuring activity, the form shall be in the size and shape specified (see 6.2).

3.4 Workmanship. The material shall be free of contamination and shall be uniform from container to container within each lot.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of tests. All quality conformance tests shall be classified, as follows:

a. Lot acceptance tests (see 4.5).
b. Periodic lot check tests (see 4.6).

4.3 Lot size. Lot size shall consist of all the molding or extrusion material manufactured at one time, from one batch, forming part of one contract or order, and submitted for acceptance at the same time and place.

4.4 Sampling. A sufficient quantity of material from one unit representative of each lot (4.3) shall be selected at random for quality conformance testing. Each container of material shall be considered as a unit of product. Failure of sample to meet the quality conformance requirements shall be cause for lot rejection.

4.5 Lot acceptance tests. Lot acceptance tests shall be made on each lot of material and shall be the basis for acceptance or rejection of the lot, except when periodic lot check tests are also required. Lot acceptance tests shall consist of the tests specified herein.

4.5.1 Melt flow. Melt flow shall be determined for nonreinforced PES materials only, in accordance with ASTM D 1238 using a load of 5000 g (including push rod) at a temperature of 360 $\pm$ 5°C.

4.5.2 Glass transition temperature. The glass transition temperature, Tg, shall be determined in accordance with ASTM D 3418.

4.5.3 Tensile strength. Tensile strength shall be determined in accordance with ASTM D 638.

4.5.4 Flammability. Unless otherwise specified (see 6.2 and 6.4), oxygen index and flammability rating tests shall be determined in accordance with ASTM D 2863 and UL 94, respectively.

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4.6 Periodic lot check tests. Periodic lot check tests shall be made on a lot of material as specified by the procuring activity (see 6.2). Periodic lot check tests shall consist of all the test methods specified in tables I, II and III (except for those test methods which are classified as lot acceptance tests in 4.5.1, 4.5.2, 4.5.3 and 4.5.4) and the additional test methods specified herein. When periodic lot check tests are made, they shall be included in the basis for acceptance or rejection of the lot.

4.6.1 Fiberglass content (type II only). The following procedure shall be used to determine the fiberglass content of the PES material:

a. Test five samples. Each sample shall have a minimum weight of 3 grams and shall be in a form suitable for weighing in an ignition crucible. Condition the sample in a previously weighed, ignited crucible in a desiccator for a minimum of 8 hours. Weigh the conditioned sample and crucible to within ± 0.005 gram.

b. Place the weighed crucibles with the test samples in a muffle furnace and raise the furnace temperature to 565 ± 10°C (1050 ± 50°F) at a rate that will not cause blowing or loss of the glass reinforcement. Heat the samples at 565 ± 10°C (1050 ± 50°F) to constant weight. Cool the ignited crucibles in a dessicator. Weigh to within ± 0.005 gram.

c. Calculate the glass content as follows:

$$\text{Glass content, \% (by weight)} = \frac{\text{Original weight - Weight loss}}{\text{Original weight}} \times 100$$

4.6.2 Impact resistance. Impact resistance shall be determined in accordance with ASTM D 3029.

5. PACKAGING

5.1 Packing. Packing shall be level A or Industrial as specified (see 6.2).

5.1.1 Level A. Unless otherwise specified, the material shall be packed in metal drums conforming to PPP-D-729, type III or type IV, in quantities of 181 kg (400 lbs), maximum.

Insofar as practical, drums shall be of uniform shape and size, with minimum cube and tare consistent with the protection required. Drums shall contain identical quantities and shall be closed in accordance with the applicable container specification. Fiber drums shall be furnished with a 4 mil thick polyethylene liner properly heat sealed.

5.1.2 Industrial. The material shall be packed in accordance with ASTM D 3951.

5.2 Marking. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129, with the exception that commercial marking in accordance with MIL-STD-1188 applies for commercial packaging only.

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6. NOTES

6.1 Intended use. The polyethersulfone (PES) materials covered by this specification are engineering thermoplastics intended for continuous use at 180°C. This thermoplastic is intended for use in various aircraft, aerospace and military applications, such as instrumentation housings, fittings, electrical connectors, switches and terminal blocks, nose cones, radomes, air ducting, obdurator bands, and as a matrix for glass fiber and carbon-graphite composites.

6.2 Ordering data.

6.2.1 Acquisition documents. Acquisition documents should specify the following:

a. Title, number and date of this specification
b. Type and class of PES material
c. Properties of PES material, if other than specified in 3.2
d. Specific form, if other than specified in 3.3
e. Flammability tests, if other than specified in 4.5.4
f. Periodic lot check tests, if required (see 4.6)
g. Level of packing required (see 5.1)

6.3 Polyether sulfones. Polyether sulfones are high temperature engineering thermoplastics which contain sulfone and ether groups in the polymer chain. All commercial products have phenylene rings in the polymer chain and are derived from poly(sulfo-1,4-phenylene-oxy-1,4-phenylene). Poly(p-phenylene) sulfone has a melting temperature of about 520°C.

6.4 Flammability test (aircraft). For aircraft applications, flammability testing in accordance with Federal Aviation Regulation 25, paragraph .853, Amendment 32 may be required.

Custodian:

Army - MR
Navy - EC
Air Force - 20

Preparing Activity:

Army - MR

Project No. 9330-0977

Review Activities:

Army - MD, GL, SM
Air Force - 14, 15, 23
DLA-GS

User Activities:

Navy - OS, MC

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TABLE I. Properties of nonreinforced polyethersulfone. 1/

Properties	Property Values	Test Method
General:		
Melt index	See 1.2 and 4.5.1	ASTM D 1238
Specific gravity	1.37 ± 0.02	ASTM D 792
Mold shrinkage	0.007 in/in	ASTM D 955
Water absorption, 24 h, max	0.5%	ASTM D 570
Mechanical:		
Tensile strength, yield, min 2/	82 MPa (12,000 psi)	ASTM D 638
Tensile elongation, ultimate, min	40-80%	ASTM D 638
Flexural strength, min	124 MPa (18,000 psi)	ASTM D 790
Flexural modulus, min	2551 MPa (370,000 psi)	ASTM D 790
Izod impact strength, min 3/ and 4/		ASTM D 256
Notched, 6.35 mm (0.250 in)	80 J/m (1.5 ft-lb/in)	
Unnotched, 6.35 mm (0.250 in)	no break	
Rockwell hardness, min	M80	ASTM D 785
Thermal:		
Thermal index, max	180°C	UL 746
Heat distortion temperature at 1.8 MPa (264 psi)	200°C	ASTM D 648
Glass transition temperature	222 ± 5°C	ASTM D 3418
Coefficient of linear thermal expansion	3.1×10^{-5} in/in/°F	ASTM D 696
Electrical:		
Dielectric strength, min	400 volts/mil	ASTM D 149
Dielectric constant at 1 MHz, max	3.5	ASTM D 150
Dissipation factor at 1 MHz, max	0.0035	ASTM D 150
Volume resistivity, min	10^{17} ohm-cm	ASTM D 257
Flammability:		
Limiting oxygen index, min	35	ASTM D 2863
Flammability rating	94V-0	UL 94

1/ All classes and grades of nonreinforced PES have the same minimum and maximum property requirements.

2/ MPa X 145 = psi

3/ J/m X 18.73 X 10^{-3} = ft-lb/in

4/ Alternative impact tests are contained in ASTM D 3029. Additional impact tests and requirements shall be specified by the procuring activity.

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TABLE II. Properties of glass fiber reinforced (GFR) polyethersulfone.

Properties	Property Values					Test Method
	Class 1 (15% GFR)	Class 2 (20% GFR)	Class 3 (30% GFR)	Class 4 (40% GFR)	Class 5 (15% PTFE/30% GFR)	
General:						
Specific gravity	1.47 ±0.02	1.51 ± 0.02	1.60 ± 0.02	1.68 ± 0.02	1.70 ± 0.02	ASTM D 792
Glass fiber content, by weight	15 ± 2%	20 ± 2%	30 ± 2%	40 ± 2%	30 ± 2%	4.6.1
Mold shrinkage	0.003 in/in	0.003 in/in	0.002-0.003 in/in	0.0015 in/in	0.0015 in/in	ASTM D 955
Water absorption, 24 hr, max	0.5%	0.5%	0.5%	0.5%	0.5%	ASTM D 570
Mechanical:						
Tensile strength, yield, min 1/	114 MPa (16,500 psi)	103 MPa (15,000 psi)	117 MPa (17,000 psi)	131 MPa (19,000 psi)	103 MPa (15,000 psi)	ASTM D 638
Tensile elongation, ultimate, min	1-5%	1-5%	1-5%	1-5%	1-5%	ASTM D 638
Flexural strength, min	158 MPa (23,000 psi)	158 MPa (23,000 psi)	172 MPa (25,000 psi)	207 MPa (30,000 psi)	158 MPa (23,000 psi)	ASTM D 790
Flexural modulus, min	4,827 MPa (700,000 psi)	5,517 MPa (800,000 psi)	7,586 MPa (1,100,000 psi)	11,034 MPa (1,600,000 psi)	5,517 MPa (800,000 psi)	ASTM D 790
Izod impact strength, min 2/ and 3/						
Notched, 6.35 mm (0.250 in)	-	75 J/m (1.4 ft-lb/in)	80 J/m (1.5 ft-lb/in)	85 J/m (1.6 ft-lb/in)	80 J/m (1.5 ft-lb/in)	ASTM D 256
Unnotched, 6.35 mm (0.250 in)	-	427 J/m (8.0 ft-lb/in)	534 J/m (10 ft-lb/in)	640 J/m (12 ft-lb/in)	347 J/m (6.5 ft-lb/in)	
Rockwell hardness, min	R118	M98	M98	M98	M98	ASTM D 785
Thermal:						
Thermal index, max	-	180°C	180°C	180°C	180°C	UL 746
Heat distortion temperature at 1.8 MPa (264 psi), min	200°C	205°C	210°C	210°C	210°C	ASTM D 648
Glass transition temperature	225 ± 5°C	225 ± 5°C	225 ± 5°C	225 ± 5°C	225 ± 5°C	ASTM D 3418
Coefficient of linear thermal expansion	2.0 X 10^{-5} in/in/°F	2.0 X 10^{-5} in/in/°F	1.8 X 10^{-5} in/in/°F	1.6 X 10^{-5} in/in/°F	1.9 X 10^{-5} in/in/°F	ASTM D 696
Electrical:						
Dielectric strength, min	380 volts/mil (14.9 MV/m)	380 volts/mil (14.9 MV/m)				ASTM D 149
Dielectric constant at 1 MHz, max	3.9	3.9				ASTM D 150
Dissipation factor at 1 MHz, max	0.007	0.007				ASTM D 150
Volume resistivity, min	10^{16} ohm-cm	10^{16} ohm-cm	10^{16} ohm-cm	10^{16} ohm-cm	10^{16} ohm-cm	ASTM D 257
Flammability:						
Limiting oxygen index, min	40	40	40	40	40	ASTM D 2863
Flammability rating	94V-O	94V-O	94V-O	94V-O	94V-O	UL 94

1/ MPa X 145 = psi

2/ J/m X 18.73 X 10-3 = ft-lb/in

3/ Alternative impact tests are contained in ASTM D 3029. Additional impact tests and requirements shall be specified by the procuring activity.

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TABLE III. Properties of carbon fiber reinforced polyethersulfone.

Properties	Property Values		Test Method
	Class 1 (30% CFR)	Class 2 (15% PTFE/30% CFR)	
General:			
Specific gravity	1.48 ± 0.02	1.57 ± 0.04	ASTM D 792
Carbon fiber content	30 ± 2%	30 ± 2%	ASTM D 3171
Mold shrinkage	-	0.001 - 0.003 in/in	ASTM D 955
Water absorption, 24 h, max	0.30%	0.25%	ASTM D 570
Mechanical:			
Tensile strength, yield, min 1/	158 MPa (23,000 psi)	128 MPa (20,000 psi)	ASTM D 638
Tensile elongation, ultimate, min	2-3%	2-3%	ASTM D 638
Flexural strength, min	234 MPa (34,000 psi)	214 MPa (31,000 psi)	ASTM D 790
Flexural modulus, min	12,066 MPa (1,750,000 psi)	11,720 MPa (1,700,000 psi)	ASTM D 790
Izod impact strength, min 2/			
Notched, 6.35 mm (0.250 in)	53 J/m (1.0 ft-lb/in)	43 J/m (0.8 ft-lb/in)	ASTM D 256
Unnotched, 6.35 mm (0.250 in)	267 J/m (5.0 ft-lb/in)	214 J/m (4.0 ft-lb/in)	
Thermal:			
Thermal index	180°C	180°C	UL 746
Heat distortion temperature at 1.8 MPa (264 psi), min	210°C	210°C	ASTM D 648
Coefficient of linear thermal expansion	0.8 X 10^{-5} in/in/°F	0.8 X 10^{-5} in/in/°F	ASTM D 696
Flammability:			
Flammability rating	94V-0	-	UL 94

1/ MPa X 145 = psi

2/ J/m X 18.73 X 10^{-3} = ft-lb/in

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-46185	PLASTIC MOLDING AND EXTRUSION MATERIALS, POLYETHERSULFONE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

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7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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